

DISSIPATION SUCCEEDS WHERE UNITARITY CANNOT: A MEAN-FIELD REALIZATION OF B_X

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ABSTRACT. [4] proves that no *unitary* generator — quadratic, quartic-diagonal (Kerr), or any natural quaternion-motivated combination of the shape modes with a second oscillator pair — realizes B_X 's classical flow, $x(\phi) = x_0 e^\phi$, $y(\phi) = y_0 e^{-\phi}$, $Y = \sqrt{xy}$ exactly constant. That result concerns closed, reversible (Hamiltonian-generated) dynamics only. This note shows that a genuinely *dissipative* process — outside the class [4] ruled out — succeeds, exactly, at the level of mean photon numbers: coupling mode 1 to a gain reservoir and mode 2 to a loss reservoir, at equal rates, via the standard Lindblad dissipators $\mathcal{D}[a_1^\dagger]$, $\mathcal{D}[a_2]$, gives $\langle n_1(\phi) \rangle = (n_1(0) + 1)e^\phi - 1$, $\langle n_2(\phi) \rangle = n_2(0)e^{-\phi}$ — B_X 's flow exactly, not approximately — with $(\langle n_1 \rangle + 1)\langle n_2 \rangle$ exactly constant. We verify this both from the closed moment equations and from direct integration of the full master equation (not assuming factorization), confirming the two modes remain in a product state throughout. We are explicit and emphatic about what this does not establish: Lindblad evolution is irreversible and non-unitary, degrades pure states into mixed ones, and is a completely different mathematical object (a superoperator on density matrices) from a Lie-algebra generator whose exponential would give a genuine quantum representation of the classical symmetry. This result does not contradict [4]; it answers a different, complementary question — is B_X 's classical flow realizable by *any* quantum process — in the affirmative, while leaving the original question — is there a unitary quantum B_X — exactly as open as [4] left it.

1. WHAT THE UNITARY NO-GO LEFT UNEXPLORED

[4] rules out closed-system, Hamiltonian-generated dynamics: single or independent-sum passive/active bilinears, and Kerr-type diagonal additions to them. All of these are *unitary* — generated by $\exp(-iHt)$ or $\exp(rK)$ for Hermitian or anti-Hermitian H, K . This leaves an entire further class of Gaussian-preserving quantum operations untouched: dissipative channels, described not by a Hamiltonian but by a Lindblad master equation, which are not reversible and do not preserve the purity of a state. We test whether this broader class succeeds where the narrower one provably cannot.

2. ENGINEERED GAIN AND LOSS

Definition 2.1. Let ρ evolve by $\dot{\rho} = \kappa \mathcal{D}[a_1^\dagger]\rho + \kappa \mathcal{D}[a_2]\rho$, where $\mathcal{D}[L]\rho := L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$: mode 1 coupled to a gain reservoir, mode 2 to a loss reservoir, at equal rates κ .

Lemma 2.2 (Exact linear moment equations).

$$(1) \quad \frac{d\langle n_1 \rangle}{dt} = \kappa(\langle n_1 \rangle + 1), \quad \frac{d\langle n_2 \rangle}{dt} = -\kappa\langle n_2 \rangle.$$

Proof. For a single Lindblad term with jump operator L , $\frac{d\langle O \rangle}{dt} = \langle L^\dagger O L \rangle - \frac{1}{2}\langle L^\dagger L O \rangle - \frac{1}{2}\langle O L^\dagger L \rangle$. For $L = \sqrt{\kappa} a^\dagger$, $O = n = a^\dagger a$: using $na^\dagger = a^\dagger(n+1)$ and $aa^\dagger = n+1$, $L^\dagger O L =$

$\kappa a n a^\dagger = \kappa(n+1)^2$; $L^\dagger LO = OL^\dagger L = \kappa a a^\dagger n = \kappa n(n+1)$. So $\frac{d\langle n \rangle}{dt} = \kappa[(n+1)^2 - n(n+1)] = \kappa(n+1)$, giving the first equation of (1). For $L = \sqrt{\kappa} a$: using $na = a(n-1)$, $L^\dagger OL = \kappa a^\dagger na = \kappa n(n-1)$; $L^\dagger LO = OL^\dagger L = \kappa a^\dagger a n = \kappa n^2$. So $\frac{d\langle n \rangle}{dt} = \kappa[n(n-1) - n^2] = -\kappa n$, giving the second equation. $\square$

Theorem 2.3 (Exact agreement with B_X). *The solution of (1), $\langle n_1(t) \rangle = (n_1(0) + 1)e^{\kappa t} - 1$, $\langle n_2(t) \rangle = n_2(0)e^{-\kappa t}$, matches B_X 's classical flow $x(\phi) = x_0 e^\phi$, $y(\phi) = y_0 e^{-\phi}$ exactly under $x = n_1 + 1$, $y = n_2$, $\phi = \kappa t$, and*

$$(2) \quad (\langle n_1(t) \rangle + 1) \langle n_2(t) \rangle = (n_1(0) + 1) n_2(0)$$

is exactly constant.

Proof. Direct substitution: $(\langle n_1 \rangle + 1) \langle n_2 \rangle = (n_1(0) + 1)e^{\kappa t} \cdot n_2(0)e^{-\kappa t} = (n_1(0) + 1)n_2(0)$, the exponentials cancelling exactly. $\square$

Numerical Verification. With $n_1(0) = 3$, $n_2(0) = 2$ (so $Y^2 = (n_1(0) + 1)n_2(0) = 8$), (1)–(2) give $Y^2 = 8.0000$ at $\phi = 0, 0.5, 1, 1.5, 2$ to displayed precision, confirmed by direct evaluation of the closed forms. We then integrated the *full* master equation (not the mean-value ODEs) on a truncated ($N = 15$ per mode) two-mode Fock space from the product Fock state $|3, 2\rangle$: $\langle (n_1 + 1)n_2 \rangle$, computed from the full density matrix, matches $(\langle n_1 \rangle + 1)\langle n_2 \rangle$ (the factorized product of separately-computed means) exactly at every sampled $t \in \{0, 0.15, 0.3, 0.45\}$, and $\text{tr } \rho = 1$ to 10^{-6} throughout. A small drift in Y^2 (from 8.0000 to 7.9984 at the latest time sampled) is attributable to the finite $N = 15$ truncation becoming inadequate as $\langle n_1 \rangle$ grows past 5, not to any breakdown of (2) itself.

Remark (No entanglement develops). The exact match between the full- ρ moment $\langle (n_1 + 1)n_2 \rangle$ and the factorized $(\langle n_1 \rangle + 1)\langle n_2 \rangle$ confirms the two modes remain in a product state throughout: gain and loss act independently on their own modes, with no interaction term to generate correlations between them.

3. WHAT THIS DOES NOT ESTABLISH

We state this plainly rather than let the clean agreement above imply more than it does.

- (i) **This is not a unitary generator.** A Lindblad master equation is a superoperator acting on density matrices, not an element of $\mathfrak{so}(2, 1)$ or any Lie algebra whose exponential gives a reversible transformation. There is no single operator $\hat{B}_X$ here whose flow this is; there is a dissipator, and dissipators are not invertible.
- (ii) **Purity is not preserved.** Starting from the pure state $|3, 2\rangle$, the evolution described here generically produces a mixed state (the gain and loss reservoirs inject genuine quantum noise). B_X 's classical flow is a well-defined trajectory on the cone; this construction reproduces it only in the sense of first moments of a decohering quantum state, not as a coherent quantum trajectory.
- (iii) **This does not contradict [4].** That note's Lemmas and Proposition concern generators of unitary evolution exclusively. Nothing here is a counterexample to any claim made there; it is a genuinely different class of quantum operation, outside the scope of what was ruled out.

Remark (What would be needed to go further). Whether *some* unitary construction — necessarily non-Gaussian and not reducible to Kerr-enhanced Gaussian coupling, per [4] —

could reproduce B_X 's flow coherently remains exactly as open as it was. What this note adds is a existence result of a different kind: the flow itself is physically achievable by a quantum process, which was not previously established in any form, coherent or not.

4. DISCUSSION

This result reuses, in a new role, exactly the Lindblad machinery already developed for the boson-bath calculations of [2]: there, gain and loss dissipators were used to explain *why* a coordinate relaxes toward an equilibrium set by a bath. Here, the same machinery is repurposed constructively: rather than asking what a bath does to an existing coordinate, we ask whether an engineered bath can be built to reproduce a specific target flow, and for B_X specifically, it can, exactly, at the mean-field level. The honest reading is that this substantially reframes rather than resolves the original question of [3]: the search for a coherent, unitary quantum B_X remains unsuccessful, but the question of whether B_X 's classical content has *any* quantum-mechanical existence now has a clear, verified, affirmative answer, of a kind distinct from what a symmetry generator would provide.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic's Claude) were used in the derivation, symbolic and numerical cross-checking, and L^AT_EX preparation of this note. This includes: the direct operator-algebra derivation and verification of Lemma 2.2's two moment equations; the closed-form verification of Theorem 2.3 and the exact cancellation in (2); the full master-equation simulation verifying the closed-form solution against direct numerical integration of the density matrix (not merely the mean-value ODEs), including the factorization check ruling out spurious inter-mode correlations; and identification of the finite-truncation origin of the small late-time drift in the numerical Y^2 . All mathematical claims were independently verified by the author.

REFERENCES

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